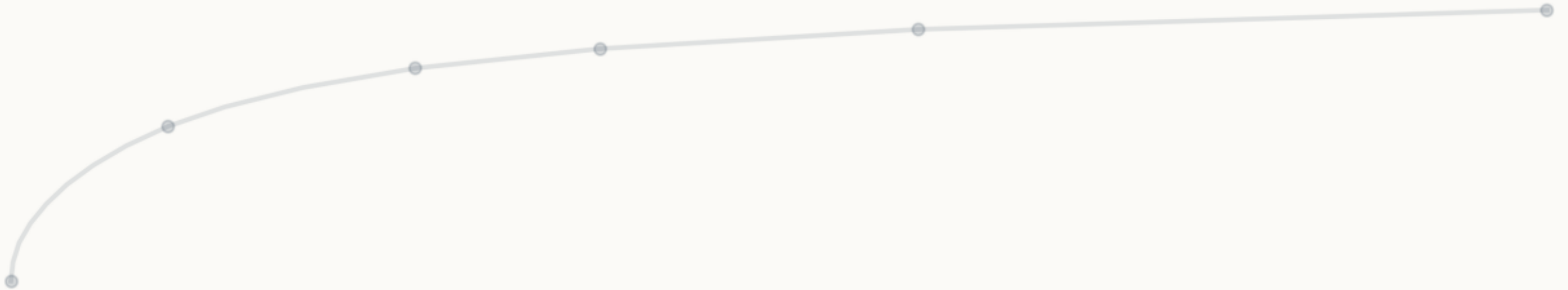


SustainaFund

A \$100 million ESG-constrained equity portfolio — efficient frontier, three mandates, and what survived out-of-sample testing

HTW Berlin Summer School 2026 · FICO Xpress case study · September 2026



The deliverable is a frontier, not a single portfolio

\$100M

budget

1,077

investable stocks

15

frontier points

Given by the brief

1% ≤ weight ≤ 20% per stock

Each region ≤ 60%

At least 30 holdings

Weighted-average ESG ≥ 70

Added by us

Sector cap 30%

Per-stock ESG floor ≥ 30

Tier 1 weapons exclusion (always on)

Tier 2 weapons exclusion (toggle, off)

Country cap 25% (built, measured, ships off)

- Risk appetite was never specified in the brief
- So we solve the whole frontier and report three points
- Every addition of ours is priced later in this deck

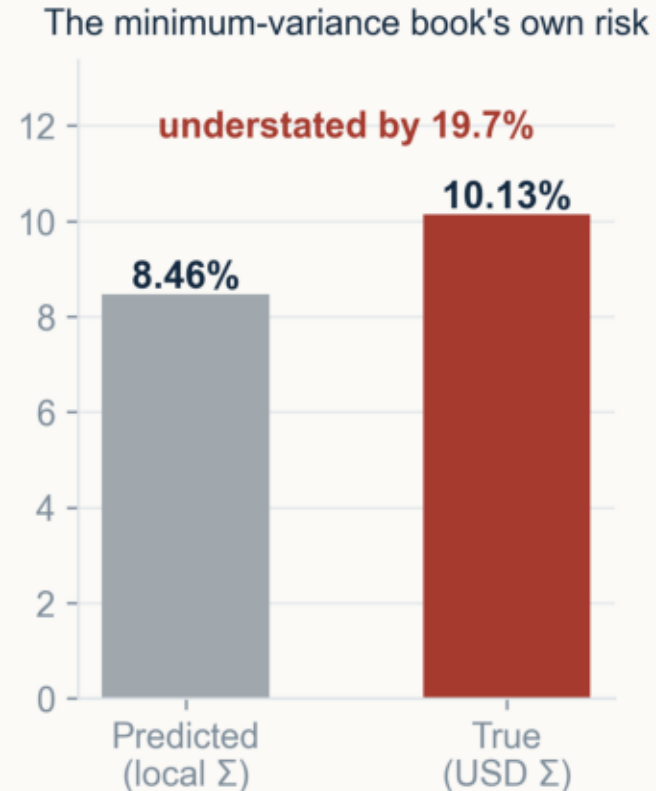
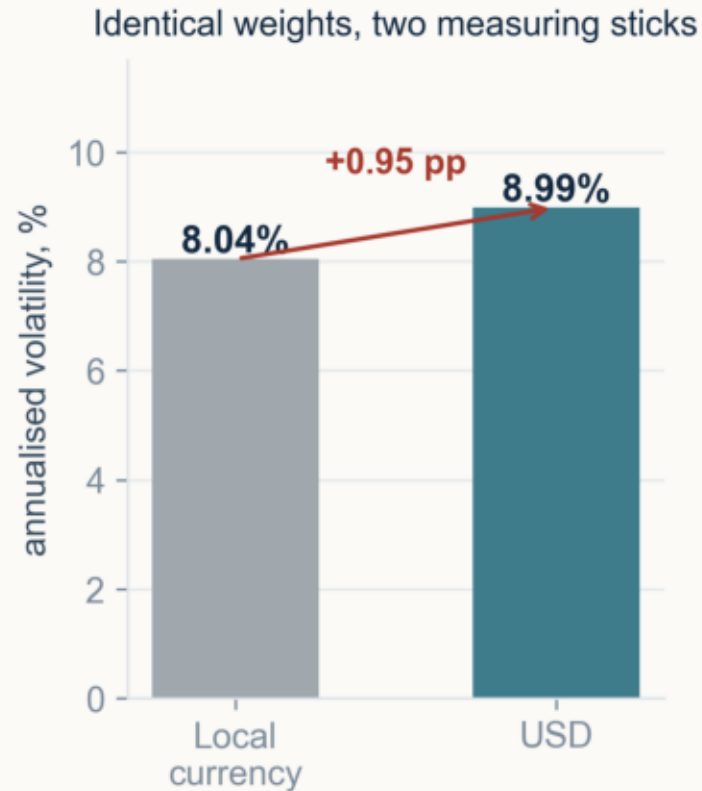
Risk appetite was never given to us, so we report the frontier and three mandates on it.

1,093 tickers in, 1,077 investable out



- Ten years of daily LSEG data, 2015 to 2025
- Converted to USD at daily ECB reference rates
- Total-return prices, not closing prices

Currency is a risk factor, not a unit conversion

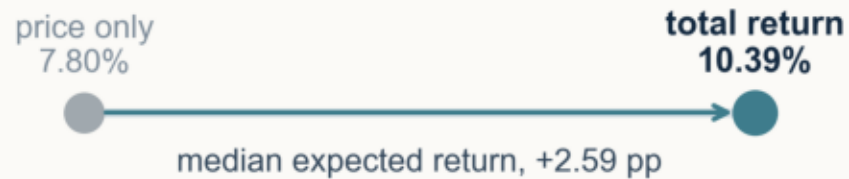


- $\ln(P_{\text{usd}}) = \ln(P_{\text{local}}) + \ln(fx)$ — the FX term does not cancel

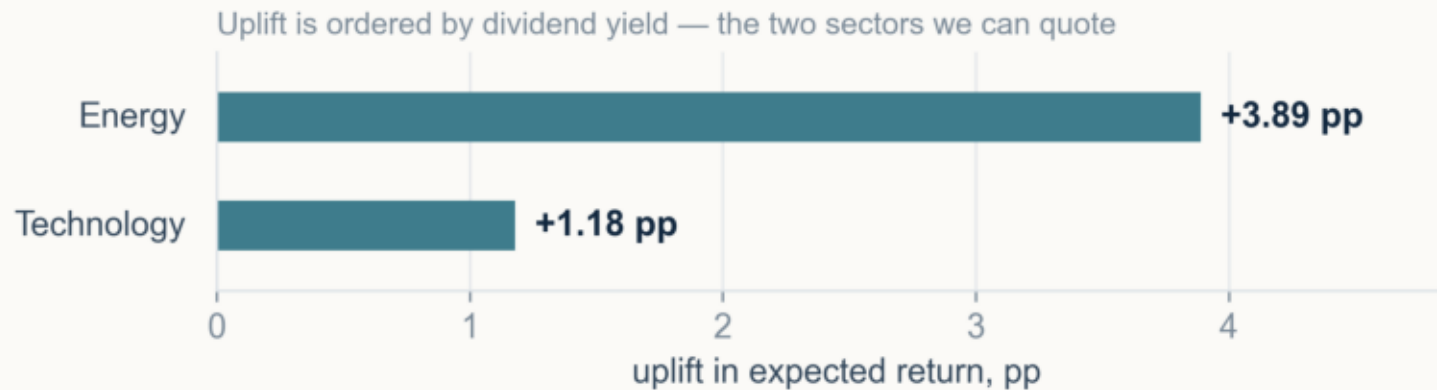
- FX is 17.6% of total portfolio variance

- A fund reports in one currency, not eight

Price returns penalise the stocks min-variance wants



- Stocks with negative expected return: 3 → 0
- Rank correlation 0.972 — levels move, order mostly holds
- We validated the vendor series with three checks



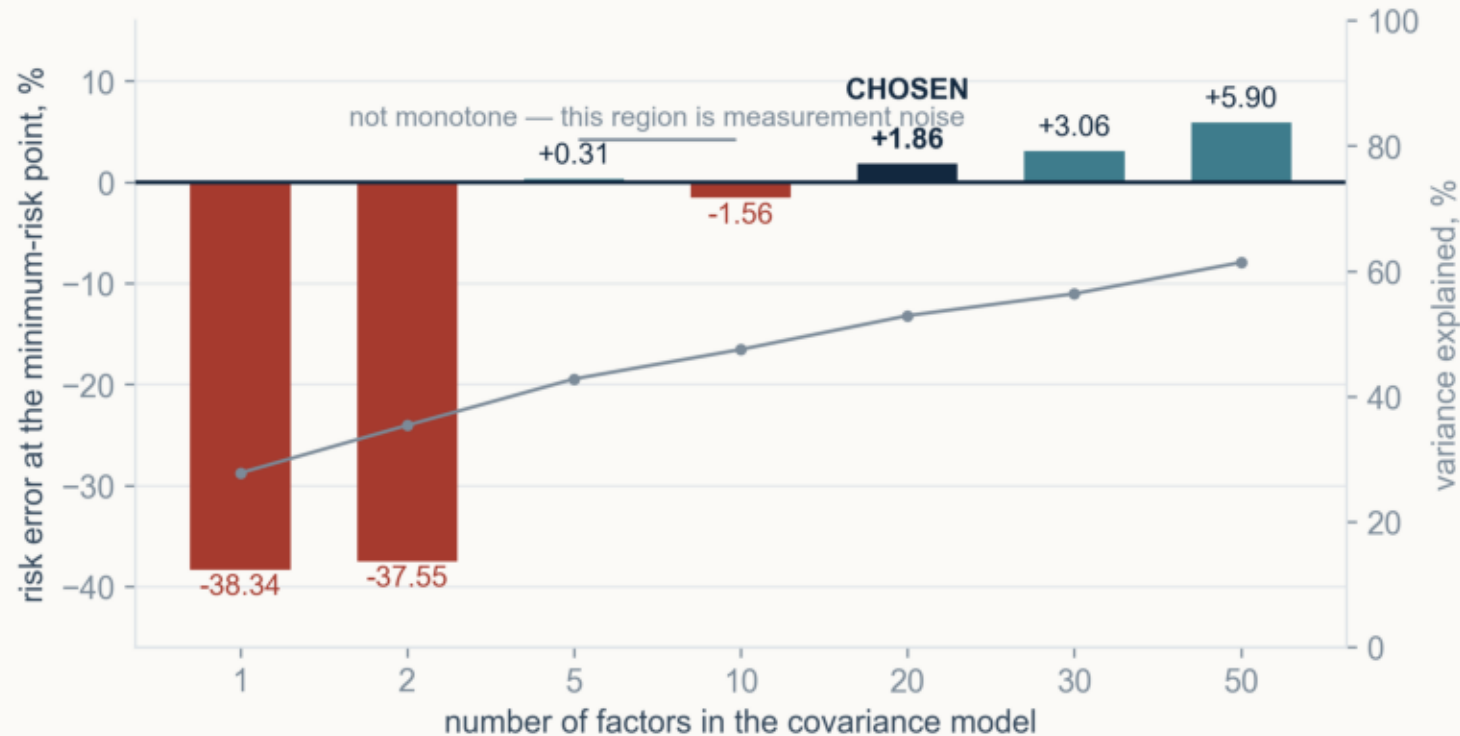
A 62.6% historical mean is not a forecast



- $w_i = \tau^2 / (\tau^2 + se_i^2)$, shrinking toward the grand mean
- Both τ^2 and se^2 are estimated from the data
- No tuning parameter anywhere in the estimator

The amount of shrinkage is measured, not chosen — there is no knob for a referee to attack.

Twenty factors, chosen by measurement, not convention



- 20 is not the smallest k that reaches zero — five is
- It is the smallest k whose margin holds either way
- At k = 50 the condition number worsens 2,957 → 4,389

We chose the smallest factor count whose margin does not depend on which portfolio you pick.

A MIQP, because the cheaper formulations failed

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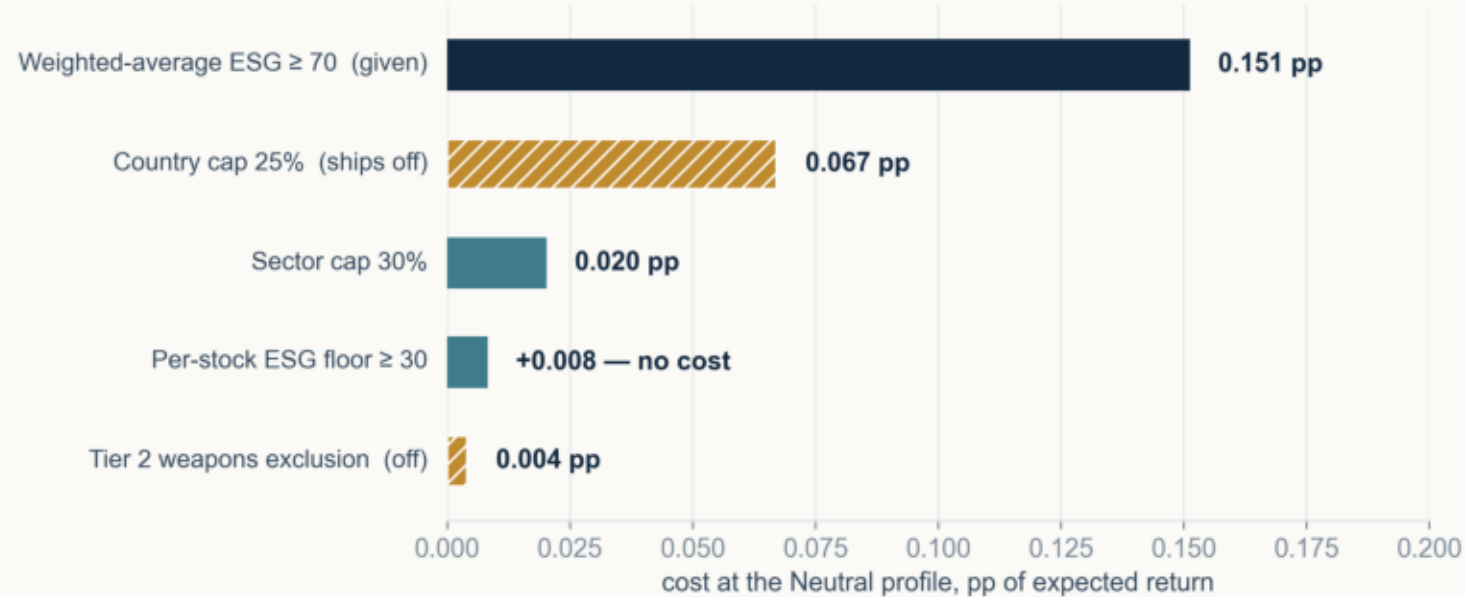
minimise    w' Σ w
subject to  w' μ ≥ β
            Σ w = 1
            0.01 y_i ≤ w_i ≤ 0.20 y_i
            y_i ∈ {0,1},    Σ y_i ≥ 30

```

Rejected	The measurement that killed it
Linear risk proxy $\sum w_i \sigma_i$	19.2% more true risk; misreports its own risk by 1.64×
Semi-continuous variables	returned 26 real positions while reporting 30
Weighted-sum λ scalarisation	binaries make the set non-convex — reaches only the convex hull; native multi-o

1,077 continuous + 1,077 binary variables · ~1.16 M quadratic elements · 0.3–1.1 s per solve · FICO Xpress 9.9.1 at a 0.1% MIP gap

Every constraint carries a price we can quote

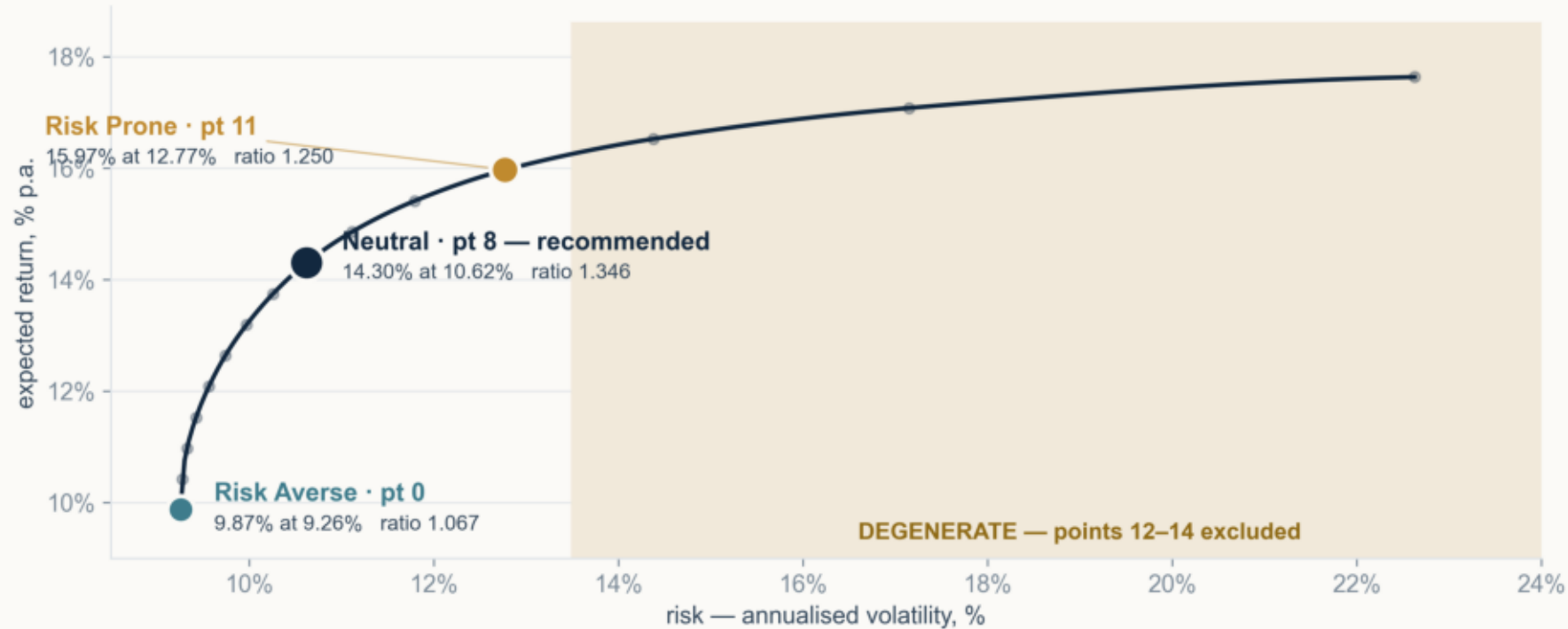


hatched = built and measured but not active. The country-cap figure is return given up at matched risk; the same cap costs 0.112 pp of risk at matched return.

- The brief's own averaging rule carries nearly all the cost
- The per-stock floor we added is effectively free
- The sector cap does not bind at Neutral — 27.10% uncapped

Rheinmetall carries an ESG score of 87.3 and would take 10.2% of the book with no explicit weapons screen.

Fifteen solved portfolios — three are mandates



curve interpolated between solved points for shape; only the six exact points are labelled

- Steep on the left, flat on the right
- Past roughly 14% risk, extra risk buys very little return

That shape is the entire argument for not choosing the right-hand corner.

Neutral is the best risk-adjusted point on the frontier

14.30%

expected return

10.62%

risk

1.346

return / risk

Profile	Return	Risk	Return/Risk	Holdings	Weighted ESG
Risk Averse (pt 0)	9.87%	9.26%	1.067	42	70.00
Neutral — recommended (pt 8)	14.30%	10.62%	1.346	30	70.00
Risk Prone (pt 11)	15.97%	12.77%	1.250	30	70.00

- Risk Averse = minimum variance
- Neutral = highest return/risk on the frontier
- Risk Prone = highest return among non-degenerate points

Neutral needs no risk-aversion parameter — it is the best point on a frontier we did not tune.

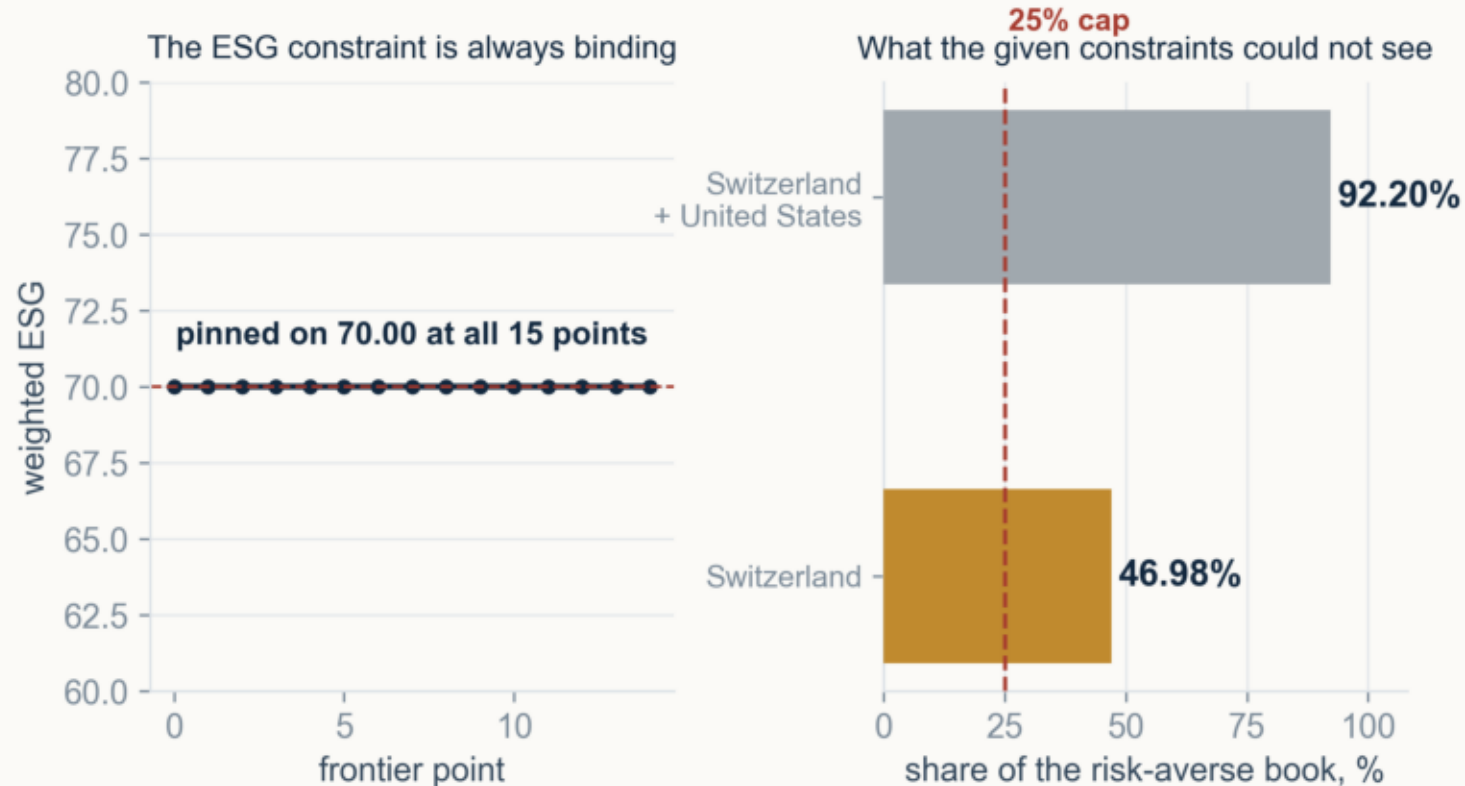
The maximum-return corner is a dominated portfolio

Point	Return	Risk	Return/Risk	Top-3 weight	At the 1% floor	Status
11	15.97%	12.77%	1.250	33.8%	10	Risk Prone
12	16.52%	14.38%	1.146	34.0%	16	degenerate — floor count
13	17.08%	17.15%	1.000	40.5%	21	degenerate — both tests
14	17.63%	22.62%	0.779	60.0%	23	degenerate — the corner

- Degenerate: top-3 above 40%, or over 15 positions at the floor
- Thresholds swept 9×9: point 11 wins for floors 10–15
- More than 15 of 30 is simply more than half the book

0.779 is worse than the risk-averse book's 1.067 — more risk for less reward per unit.

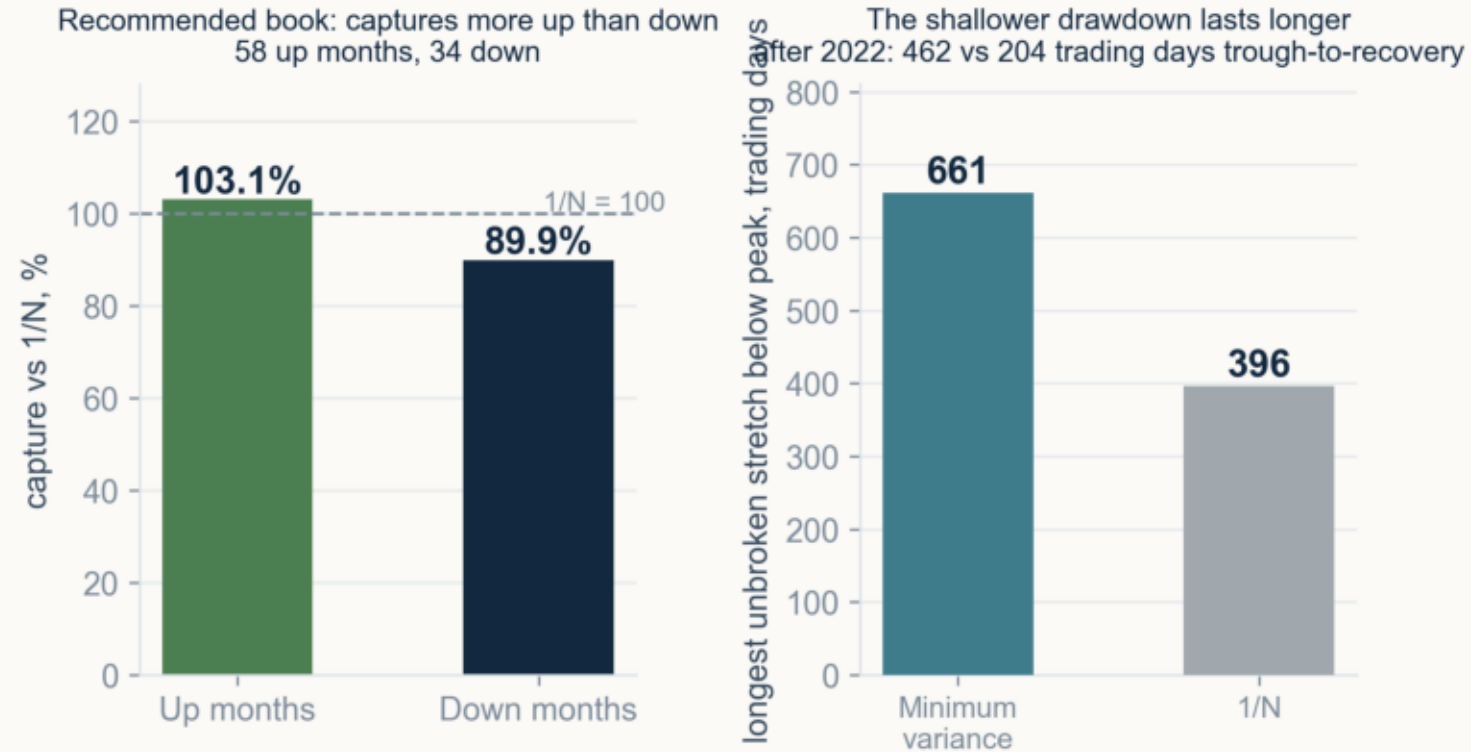
ESG binds everywhere, and the constraints missed Switzerland



- A region cap cannot see one country
- A sector cap cannot see a country spanning two sectors
- A 25% cap costs 0.067 pp and removes 11.7 pp of concentration

The US exemption is forced, not chosen — with a 25% cap the infeasibility is exactly $1 - 0.60 - 0.25 = 0.15$.

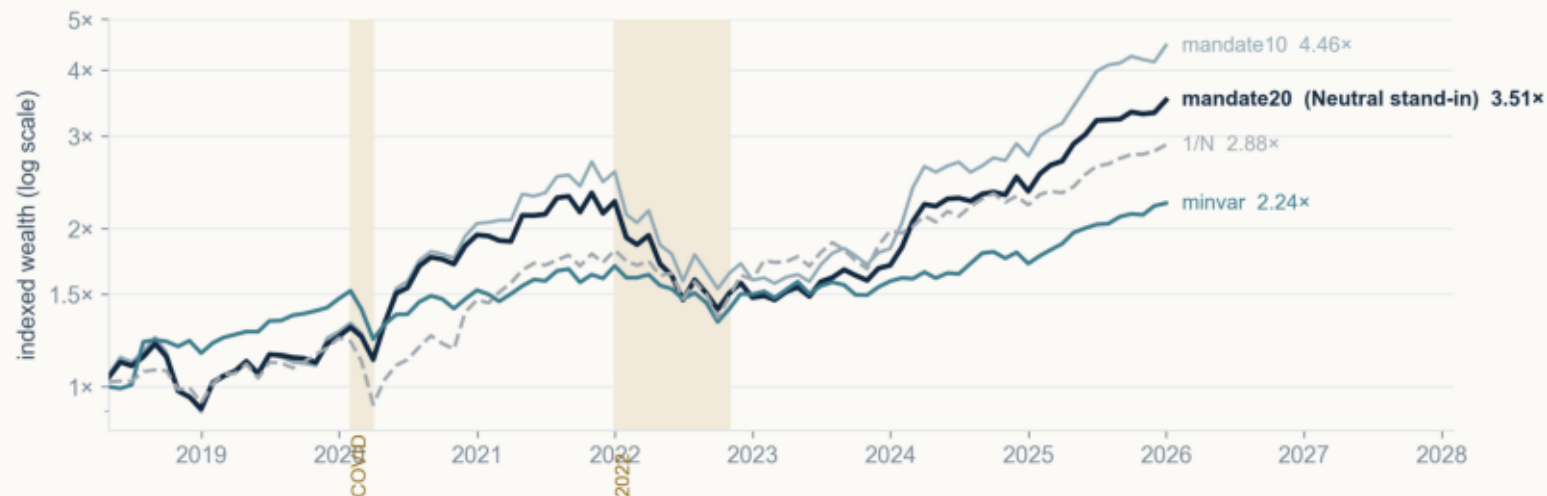
Averages hide what the portfolio actually does



- Captures 103% of up months, 90% of down months
- Minimum variance stays underwater 661 days vs 396 for 1/N
- The shallower drawdown is also the longer one

The capture asymmetry is the strongest argument for the recommendation, and a Sharpe ratio cannot show it.

Out of sample: risk edge real, return edge not

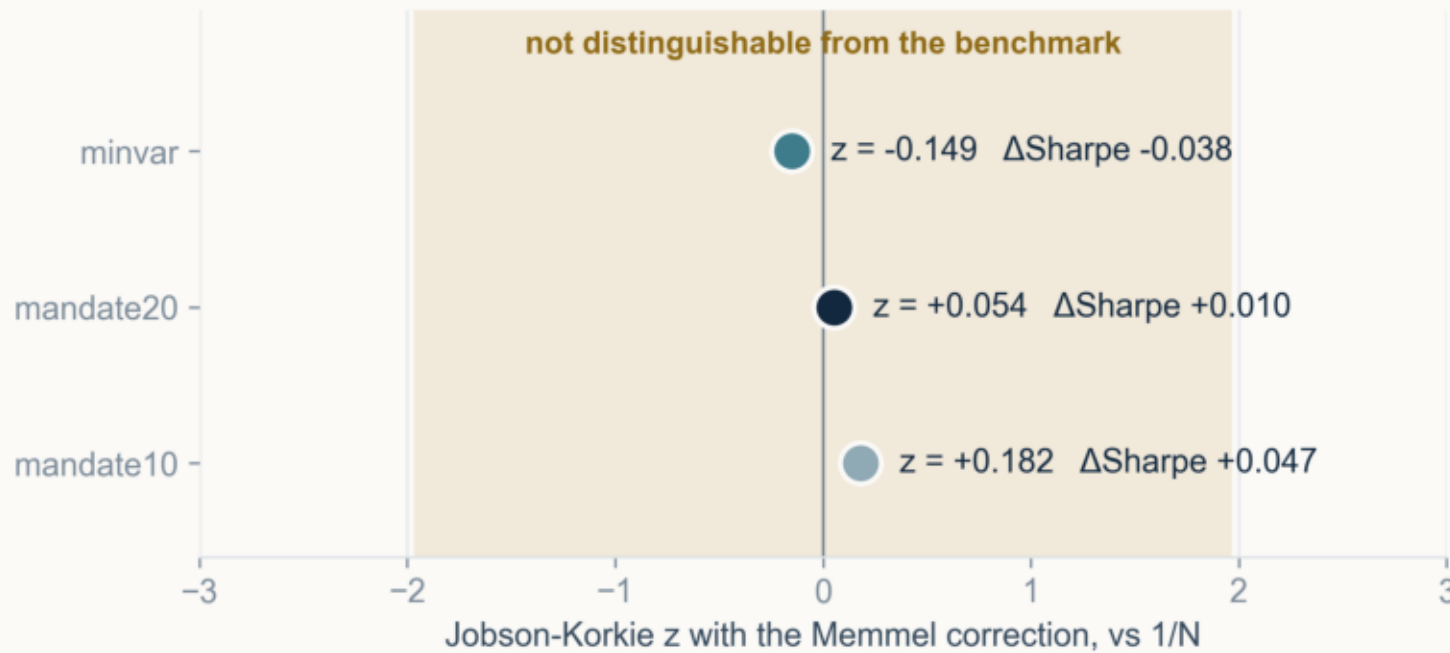


Book	CAGR	Volatility	Sharpe	Max DD
minvar	10.94%	13.20%	0.829	-33.81%
mandate20 (Neutral stand-in)	17.60%	20.08%	0.876	-42.31%
mandate10	21.29%	23.30%	0.914	-46.00%
1/N benchmark	14.65%	16.91%	0.866	-37.68%

- 31 rebalances, 3-year rolling window, 7.7 years
- Only data available before each date is used
- The profile is a relative mandate, comparable across regimes

Higher CAGR came with proportionally higher risk — the ranking on Sharpe barely moves.

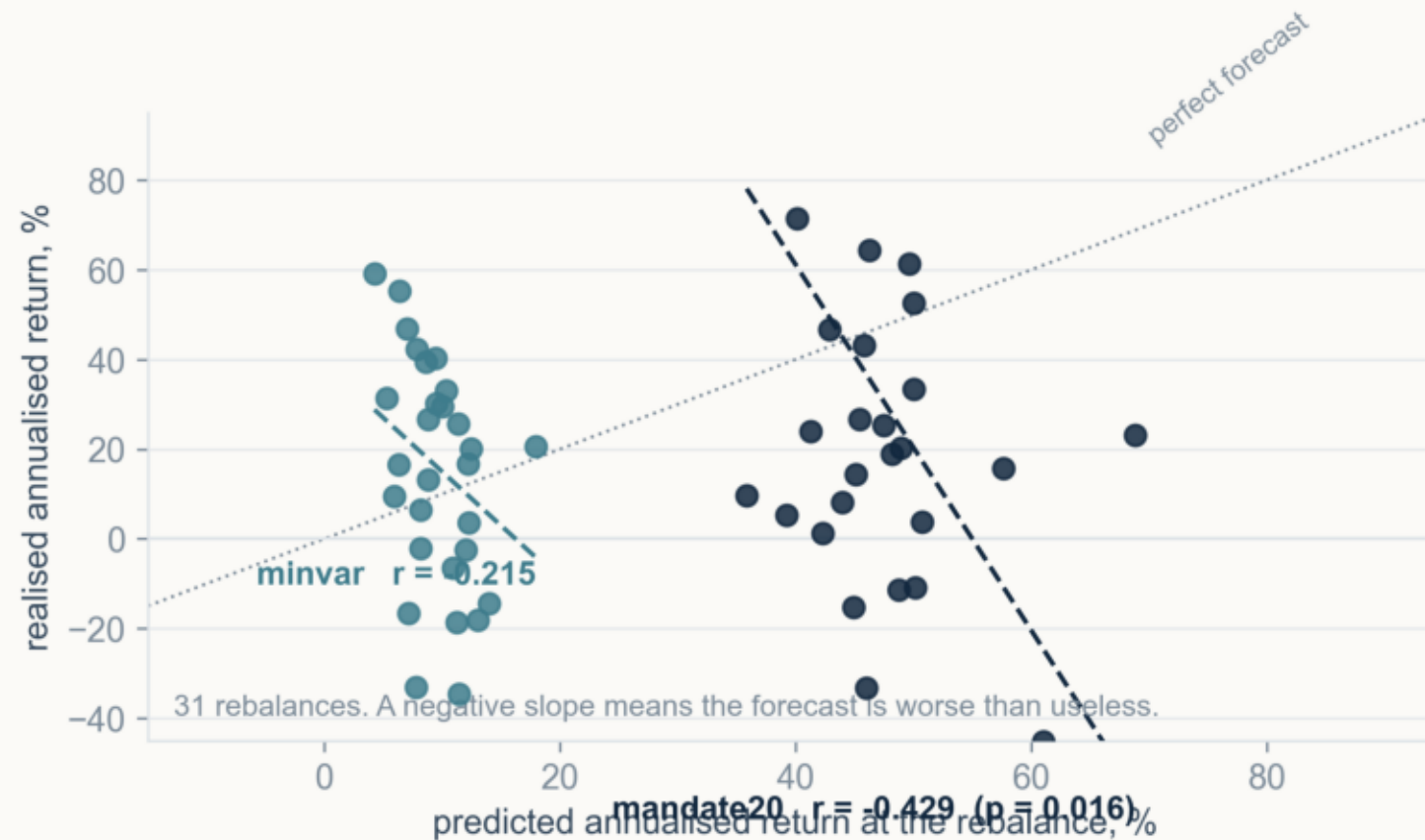
Every Sharpe difference sits inside the noise band



- After 10 bp of costs the return-seeking book falls below 1/N
- 0.857 against 0.866, at 170% annual turnover
- Jobson-Korkie with Memmel, not a t-test on returns

Over 7.7 years the honest answer is that no book is statistically distinguishable from the benchmark — in either direction.

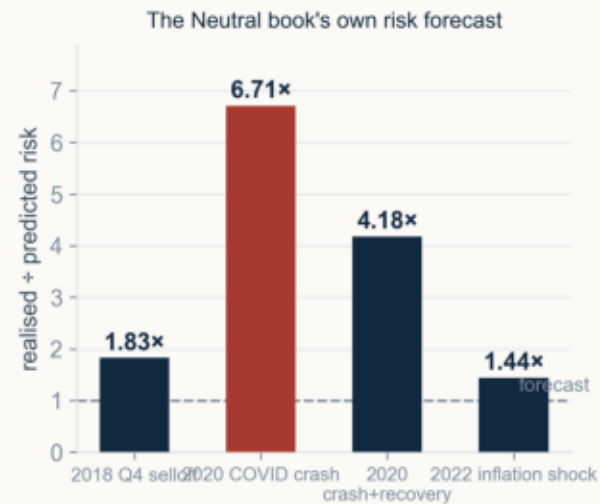
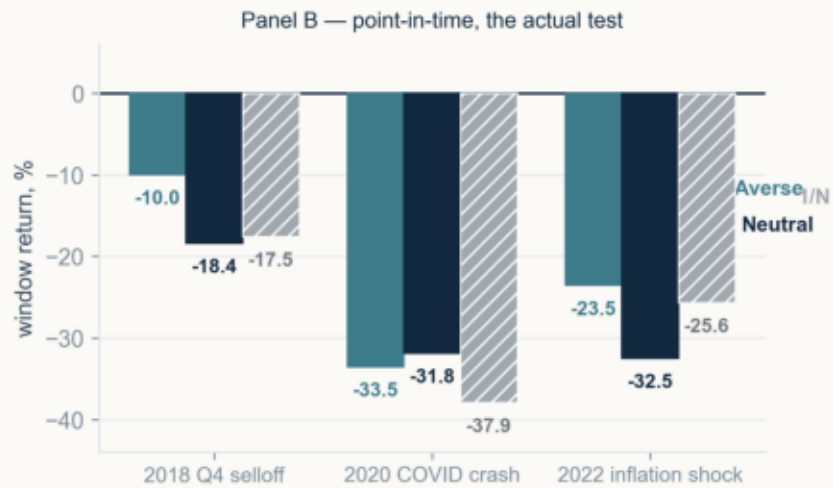
The expected-return model forecasts backwards



- Correlation -0.43 over 31 rebalances, $p = 0.016$
- Predicted 47.3%, realised 31.5%
- Minimum variance predicted 9.7%, realised 15.6%

That is error maximisation made visible, and it is why the edge is in risk and not in return.

The crash realised 6.7× the risk we forecast



- Panel A is in-sample, wins 7 of 7 — not evidence
- Panel B re-estimates on the three years before each window
- Risk Averse wins 3 of 3 crises, Neutral 1 of 3

Any risk number this model produces has to be presented as a lower bound.

Recommend on the estimate, recommend risk on the evidence

What we recommend

- Neutral, point 8
- 14.30% return at 10.62% risk
- ratio 1.346, 30 holdings, ESG exactly 70.00
- 103% up capture against 90% down

What we can demonstrate

- Volatility 13.20% vs 16.91% — a 21.9% cut
- Lower in 5 of 6 sub-periods
- Drawdown -33.81% vs -37.68%
- Beats 1/N in all three crisis windows
- — all at the minimum-variance end

What we cannot claim

- A statistically significant risk-adjusted edge
- Every Jobson-Korkie z inside ± 1.96
- Over 7.7 years and 31 rebalances
- In either direction

- ESG scores and sectors are a single present-day snapshot — a 2018 rebalance uses 2025 information
- The universe is today's index constituents, so every strategy including the benchmark looks optimistic
- Transaction costs are measured but not priced inside the optimiser

The defensible claim is about risk, not return — and stating that is stronger than a Sharpe ratio that fails a t -test.

Backup

Detail held back for questions

Factor selection, the full table

k	Variance explained	Median R ²	Risk error at min-risk
1	27.8%	0.273	-38.34%
2	35.4%	0.355	-37.55%
5	42.8%	0.436	+0.31%
10	47.5%	0.469	-1.56%
20	52.9%	0.518	+1.86%
30	56.4%	0.547	+3.06%
50	61.4%	0.586	+5.90%

- Negative means the model understates portfolio risk
- k = 20 is the chosen row
- In-sample consistency check, not a forecast test

The first four factors are economically readable

Factor	Variance	Reading	Evidence
F1	27.8%	broad market	correlation 0.996 with the equally weighted index
F2	7.5%	Europe vs United States	beta +1.43 Europe / -1.73 US
F3	3.4%	Energy vs Technology	beta -2.76 Energy / +1.75 Technology
F4	2.4%	Technology vs Utilities	beta -1.47 Technology / +2.94 Utilities

- Nothing about these was specified in advance
- They are eigenvectors of the return covariance
- An explicit region factor moved the error 13.6% → 13.1% only

Shorter estimation windows look better and are worse



- 10y Jaccard 1.00 by construction; 5y 0.283; 3y 0.193

- 3y weight correlation with the 10y book: –0.019

- μ 's range widens from 2.7–19.1% to –9.8–46.1%

Why 31 rebalances and not two

Split	Book	Δ Sharpe vs 1/N	Verdict
2019-12 / 2022-12	mandate20	+0.087	beats 1/N
2020-06 / 2023-06	mandate20	-0.244	loses to 1/N
three-point variant	mandate20	-0.130	loses to 1/N
31 rebalances	mandate20	+0.010	not distinguishable

- A spread of 0.331, decided by where the boundary falls
- Two observations cannot separate skill from luck
- We tested the objection rather than asserting it

Six reproducibility gates, and three defects they caught

Gate	Result
rewriting the group caps in a different idiom	byte-identical LP file; all 15 frontier points re-solved with max $ \text{delta weight} = 0.000$
rebuilding the covariance at $k=20$	reproduces the shipped matrix to $1.8\text{e-}15$
re-estimating on the ten-year window	reproduces μ to $1.7\text{e-}16$ and Σ to $1.8\text{e-}15$
new backtest harness vs the original	Sharpe 0.829 vs 0.835, volatility 13.197% vs 13.210%
new $1/N$ vs an independently built one	Sharpe 0.867 vs 0.866
regenerated presentation inputs	every field matches the original; the rebuilt PDF is text-identical

- A number that cannot be reproduced is not evidence
- The deck itself was unbuildable by anyone else at one point
- A $1\text{e-}6$ tolerance counted zero positions where there was one

Weapons screening, and a ticker that would have hidden it

- Tier 1: weapons are the core business — always excluded
- Tier 2: diversified defence — a toggle, currently off, costs 0.004 pp
- Four of ten Tier 2 tickers were absent from the dataset

BA.L was the dangerous one. BA.N exists in this dataset and is Boeing, so a plausible-looking correction would have excluded the wrong company.

What the 1/N benchmark is, and is not

- 0.093% per name against the brief's 1% floor
- Rebalancing frequency moves it 0.9 pp of CAGR
- The ESG-screened variant breaks the 60% region cap

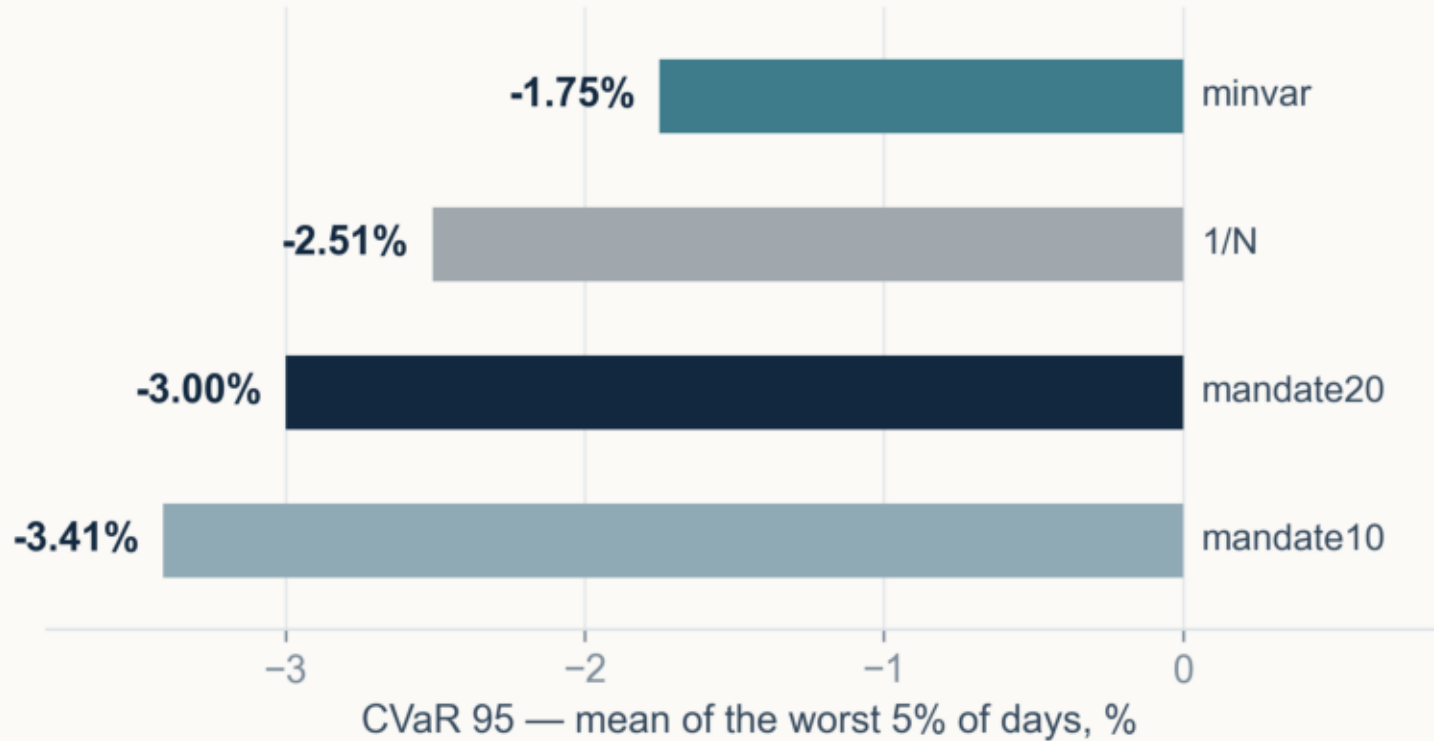
No 1/N variant is admissible under the brief. It is a reference for what no skill would have earned, not a portfolio we could have proposed.

The sub-period record across six regimes

Period	Regime	minvar CAGR	1/N CAGR	minvar vol	1/N vol
2018-04..2019-12	calm market	+24.72%	+13.04%	14.15%	11.87%
2020 full year	crash and rebound	+3.82%	+18.09%	23.44%	30.78%
2020-02..2020-03	the crash itself	-87.48%	-94.20%	63.09%	66.91%
2021	boom	+11.25%	+24.50%	8.02%	11.47%
2022	inflation shock	-12.06%	-12.60%	13.61%	21.25%
2023-2025	AI rally	+14.30%	+21.48%	8.22%	12.19%
FULL	whole test period	+10.94%	+14.65%	13.20%	16.91%

- Volatility lower in 5 of 6 sub-periods
- It wins the 2020 crash and the 2022 inflation shock
- It gives up the 2021 boom and the AI rally

Tail risk, where the minimum-variance book earns its keep



- CVaR 95 of -1.75% against -2.51% for 1/N
- The mean of the worst 5% of days
- The return-seeking books are worse than the benchmark